

TideWAL: Timestamp-Integrated Dependency-Closed Parallel WAL for Main-Memory Transaction Processing

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概要: This paper proposes TideWAL, a dependency-closed parallel WAL protocol for main-memory transaction processing. TideWAL avoids conservative global-prefix durable acknowledgment while preserving recovery safety by acknowledging a transaction only after both its own log record and the durable frontier of its observed dependencies are persistent. In an ERMIA-style implementation, TideWAL reaches 553K durable acknowledgments per second on YCSB-B at 32 worker threads, $2.3\times$ P-WAL throughput, while keeping pending durable commits to 58.

1. Introduction

1.1 Motivation

Transaction processing is a core abstraction behind data-intensive applications such as banking, payment processing, inventory management, ticket reservation, and online services that update shared state under high concurrency. A transaction processing system must provide both correct concurrent execution and crash recovery; in this sense, transaction processing consists of concurrency control (CC) and recovery [1]. A large body of work has studied scalable CC for multicore main-memory databases, including Silo, ERMIA, TicToc, SI, SSI, and SSN-style protocols [2], [3], [4], [5], [6], [7]. These protocols improve isolation and scalability by reducing centralized validation, timestamp, and version-management bottlenecks. In contrast, the recovery side of modern CC protocols has been explored less systematically: when a high-performance CC protocol is combined with crash durability, it is often unclear which recovery protocol preserves both safety and scalability.

Write-ahead logging (WAL) is the standard foundation for database recovery, with ARIES being the classical design [8], [9]. Modern systems have revisited WAL for multicore and fast-storage environments. SiloR paral-

lelizes logging, checkpointing, and recovery for Silo; Passive Group Commit reduces synchronization overhead on non-volatile memory; and P-WAL partitions the log into multiple streams [10], [11], [12]. Although these systems differ in the target storage device, log format, and recovery procedure, they share an important lesson: writing log records in parallel is essential for making durability compatible with multicore transaction processing. At the same time, strong multiversion protocols such as SI, SSI, and SSN introduce timestamp and dependency structures that are not fully reflected in conventional parallel logging designs. The motivation of this work is therefore to clarify how P-WAL-style parallel logging should be combined with such timestamp-based CC protocols, especially SSN, without unnecessarily reintroducing global ordering and durable-prefix bottlenecks.

1.2 Problem

We target SSN because it is the most complex of the SI-family protocols we consider: it certifies serializability using dependency metadata, while the underlying engine still uses timestamped multiversion execution. As the baseline recovery protocol, we start from P-WAL, which is also used as a foundation of Shirakami, the transaction processing mechanism of Tsurugi [12], [13]. P-WAL is attractive because it removes the single-WAL insertion bottleneck by assigning transactions to multiple log

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streams. However, combining P-WAL with SSN exposes three problems.

P1. Multiple Global Counter Accesses: First, P-WAL can introduce frequent global-counter accesses. Even if the CC layer already assigns a commit timestamp for SSN validation and serialization, the recovery layer may allocate a separate global LSN or maintain a global durable prefix. This duplicates the ordering work already performed by the CC layer and adds shared-memory synchronization to the commit path.

P2. Waiting for Data Transfer: Second, workers can spend a long time waiting around `fdatsync`. P-WAL parallelizes log streams, but a transaction cannot be acknowledged until the corresponding durable condition is satisfied. If worker threads wait directly for log flushes or durable-prefix updates, storage synchronization and notification latency reduce the effective scalability of the CC protocol.

P3. Total History: Third, global-prefix acknowledgment waits for log records that are unrelated to the transaction being acknowledged. This rule is safe, but overly conservative: if one WAL shard is slow, transactions on other shards may be delayed even when they have no dependency on transactions logged by the slow shard. The opposite design point, local-only acknowledgment, avoids this delay but is unsafe. If transaction T reads a value written by transaction U , acknowledging T before U is recoverable may allow a crash recovery state that contains T without the state on which T depends.

Figure 1 summarizes these alternatives. Global-prefix acknowledgment is safe but may wait for unrelated slow shards. Local-only acknowledgment avoids such waits but can acknowledge a transaction before its dependencies are recoverable. The desired point is dependency-closed acknowledgment: acknowledge a transaction after its own log record and the transactions it depends on are durable.

In asynchronous durability protocols, logical commits and durable commit acknowledgments must be distinguished. A worker may continue after enqueueing a log record, but the transaction is not durable from the client’s perspective until the system returns a durable acknowledgment. Therefore, this paper uses *durable-acknowledgment throughput* as the main throughput metric and reports it together with pending durable commits and tail latency.

1.3 Approach

This paper proposes *TideWAL*, a timestamp-integrated,

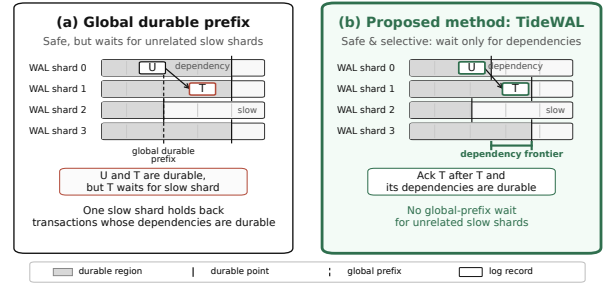


FIG 1 Durable acknowledgment policies in parallel WAL. Global-prefix acknowledgment is safe but can delay transactions whose dependencies are already durable because it waits for unrelated slow shards. Local-only acknowledgment avoids this delay but is unsafe. TideWAL uses dependency-closed acknowledgment, acknowledging a transaction only after its own log record and dependency frontier are durable.

dependency-closed parallel WAL protocol for main-memory transaction processing. TideWAL targets timestamp-based engines such as ERMIA, where the transaction engine already computes a logical commit order. Its goal is not merely to maximize raw logical commit throughput, but to return durable commit acknowledgments safely without unnecessary global-prefix waiting.

TideWAL addresses the three problems above with three corresponding techniques.

A1. Commit timestamps as logical LSNs. TideWAL reuses the SSN/ERMIA commit timestamp as the logical LSN for recovery. WAL shards still maintain local physical positions, but the WAL layer does not allocate an additional global logical order. This removes duplicated global ordering between CC and recovery and eliminates WAL-side separate global LSN allocation on the commit path.

A2. Worker/flusher/committer separation. TideWAL decouples transaction execution from durable acknowledgment. Workers execute and validate transactions, assign commit timestamps, and enqueue log records. Flushers batch and persist shard-local logs. Committers observe durable-frontier advances and return durable acknowledgments. This pipeline keeps workers from directly blocking on `fdatsync` or on durable-prefix notification waits.

A3. Dependency-closed durable acknowledgment. TideWAL replaces global-prefix acknowledgment with dependency-closed acknowledgment. For each transaction, TideWAL maintains a dependency frontier that summarizes the log positions that must be recoverable before the transaction can be acknowledged. A transaction is

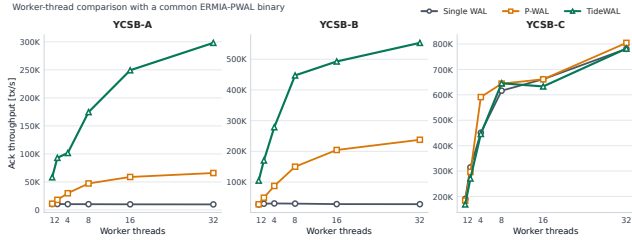


図 2 Durable-acknowledgment throughput by transaction worker threads. TideWAL improves update and mixed read-write workloads by combining parallel WAL logging, dependency-closed durable acknowledgment, and timestamp-integrated logical ordering.

acknowledged only when its own log record and this dependency frontier are durable. This preserves recovery safety while avoiding waits for unrelated WAL shards.

We implement TideWAL in an ERMIA-style engine and evaluate it using YCSB workloads with the transaction worker count on the x-axis. The comparison uses the same ERMIA-PWAL binary for all WAL systems; Single WAL and P-WAL are selected by runtime WAL-mode flags, while TideWAL additionally runs background flusher/logger and committer threads. At 32 worker threads on YCSB-A, Single WAL achieves 10K durable acknowledgments per second and P-WAL achieves 66K, whereas TideWAL achieves 298K durable acknowledgments per second. On YCSB-B, Single WAL achieves 28K and P-WAL achieves 238K, whereas TideWAL achieves 553K durable acknowledgments per second. These results correspond to $4.5\times$ and $2.3\times$ improvements over P-WAL on YCSB-A and YCSB-B, respectively. TideWAL also keeps the number of pending durable commits small at 32 worker threads: 703 pending commits on YCSB-A and 58 pending commits on YCSB-B.

Figure 2 shows the worker-thread comparison. The results show that TideWAL improves update and mixed read-write workloads where durable logging remains on the critical path. We also evaluate YCSB-C, where all transactions are read-only and WAL records are skipped. In this case, durability work largely disappears; after using the same binary and a read-only empty-frontier fast path, Single WAL, P-WAL, and TideWAL perform similarly at 32 worker threads. We therefore use YCSB-C as a sanity check for read-only bookkeeping overhead rather than as evidence of WAL persistence performance.

TideWAL’s timestamp integration is primarily a design simplification: it removes duplicated logical ordering between concurrency control and WAL durability by reusing ERMIA’s commit timestamp as the logical re-

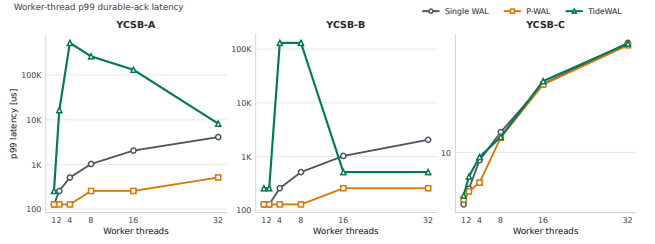


図 3 P99 durable-acknowledgment latency by transaction worker threads.

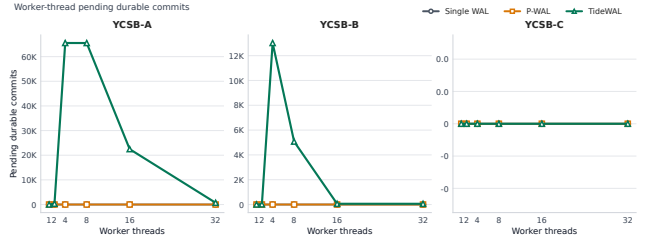


図 4 Pending durable commits by transaction worker threads.

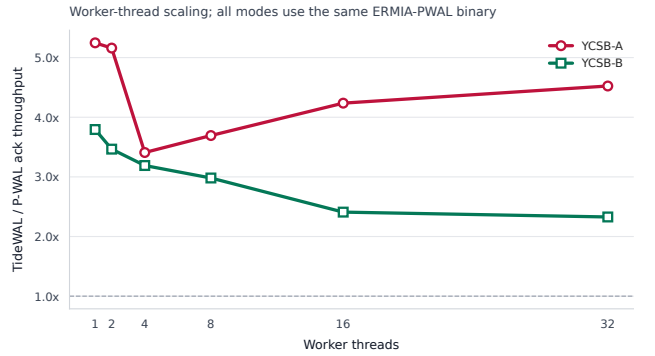


図 5 TideWAL throughput relative to P-WAL by transaction worker threads.

covery order. We do not rely on timestamp integration alone as a direct throughput optimization; the main performance effect comes from parallel logging, dependency-closed durable acknowledgment, and separating workers from durable-wait processing.

1.4 Organization

The rest of this paper is organized as follows. Section 2 reviews WAL, parallel WAL, timestamp-based concurrency control, and dependency requirements for recovery. Section 3 presents TideWAL’s timestamp-integrated logical ordering, dependency-frontier construction, and worker/flusher/committer pipeline. Section 4 evaluates durable-acknowledgment throughput, pending durable commits, and tail latency. Section 5 discusses related work. Section 6 concludes the paper.

2. Preliminaries

2.1 Write Ahead Logging

2.2 P-WAL

P-WAL is attractive because it removes the single-WAL insertion bottleneck by assigning transactions to multiple log streams. However, combining P-WAL with SSN exposes three problems.

P1. Multiple Global Counter Accesses: First, P-WAL can introduce frequent global-counter accesses. Even if the CC layer already assigns a commit timestamp for SSN validation and serialization, the recovery layer may allocate a separate global LSN or maintain a global durable prefix. This duplicates the ordering work already performed by the CC layer and adds shared-memory synchronization to the commit path.

P2. Waiting for Data Transfer: Second, workers can spend a long time waiting around `fdatasync`. P-WAL parallelizes log streams, but a transaction cannot be acknowledged until the corresponding durable condition is satisfied. If worker threads wait directly for log flushes or durable-prefix updates, storage synchronization and notification latency reduce the effective scalability of the CC protocol.

P3. Total History: Third, global-prefix acknowledgment waits for log records that are unrelated to the transaction being acknowledged. This rule is safe, but overly conservative: if one WAL shard is slow, transactions on other shards may be delayed even when they have no dependency on transactions logged by the slow shard. The opposite design point, local-only acknowledgment, avoids this delay but is unsafe. If transaction T reads a value written by transaction U , acknowledging T before U is recoverable may allow a crash recovery state that contains T without the state on which T depends.

3. Design

4. Evaluation

Table 1 summarizes the main 32-worker end-to-end results used in the figures. Table 2 extracts the TideWAL-vs-P-WAL speedups from the same measurements.

Table 3 summarizes the 32-worker perf-stat analysis. Table 4 shows the worker-scaling perf-stat data for YCSB-B, where TideWAL’s throughput improvement is most visible.

Finally, Table 5 reports the top self-time symbols for YCSB-C. Since YCSB-C is read-only, this table is used as

表 1 End-to-end YCSB results at 32 transaction worker threads.

Workload	System	Ack tps	p99 μ s	Pending	fdatasync
YCSB-A	Single WAL	9.9K	4096	0	39582
YCSB-A	P-WAL	65.9K	512	0	263401
YCSB-A	TideWAL	298.1K	8192	703	82107
YCSB-B	Single WAL	28.0K	2048	0	45012
YCSB-B	P-WAL	237.6K	256	0	381791
YCSB-B	TideWAL	553.2K	512	58	119651
YCSB-C	Single WAL	780.9K	41	0	0
YCSB-C	P-WAL	804.5K	40	0	0
YCSB-C	TideWAL	782.0K	41	0	0

Ack tps is durable-acknowledgment throughput. TideWAL uses 7 flusher threads and 1 committer thread at 32 workers.

表 2 TideWAL throughput relative to P-WAL at 32 transaction worker threads.

Workload	P-WAL tps	TideWAL tps	Speedup	TideWAL p99 μ s
YCSB-A	65.9K	298.1K	4.52×	8192
YCSB-B	237.6K	553.2K	2.33×	512
YCSB-C	804.5K	782.0K	0.97×	41

表 3 Perf-stat summary at 32 transaction worker threads.

Workload	System	Ack tps	CPU cores	Cycles/tx	Instr/tx
YCSB-A	Single WAL	9.6K	1.2	316310	648780
YCSB-A	P-WAL	65.2K	19.5	760326	447345
YCSB-A	TideWAL	295.7K	25.7	216136	144297
YCSB-B	Single WAL	27.1K	1.1	98355	157182
YCSB-B	P-WAL	246.6K	18.6	191636	108816
YCSB-B	TideWAL	542.5K	31.1	145051	63165
YCSB-C	Single WAL	791.5K	40.0	131145	28270
YCSB-C	P-WAL	810.8K	40.0	128225	28264
YCSB-C	TideWAL	772.3K	40.2	135099	32551

表 4 YCSB-B perf-stat worker scaling.

Workers	Single tps	P-WAL tps	TideWAL tps	Single CPU	P-WAL CPU
1	28.9K	29.3K	122.4K	0.5	0.5
2	32.6K	53.9K	195.1K	0.6	0.6
4	32.5K	97.7K	298.0K	0.6	0.6
8	32.6K	161.8K	452.1K	0.6	0.6
16	29.5K	205.7K	510.4K	0.8	0.8
32	29.0K	247.4K	570.6K	1.0	1.0

a sanity check that the remaining cost is read-path book-keeping rather than WAL persistence.

5. Related Work

6. Conclusion

謝辞

表 5 Top self-time symbols for YCSB-C at 32 transaction worker threads.

System	Rank	Self %	Symbol	
Single WAL	1	58.08	TxExecutor::read	[9]
Single WAL	2	23.07	TxExecutor::ssn_parallel_commit	
Single WAL	3	4.22	MasstreeWrapper[Tuple::get_value	
Single WAL	4	3.17	TxExecutor::mainte	
Single WAL	5	2.74	TxExecutor::read_internal	[10]
P-WAL	1	58.93	TxExecutor::read	
P-WAL	2	22.93	TxExecutor::ssn_parallel_commit	
P-WAL	3	3.93	MasstreeWrapper[Tuple::get_value	
P-WAL	4	3.13	TxExecutor::mainte	
P-WAL	5	2.81	TxExecutor::read_internal	
TideWAL	1	55.75	TxExecutor::read	
TideWAL	2	21.67	TxExecutor::ssn_parallel_commit	[11]
TideWAL	3	4.08	MasstreeWrapper[Tuple::get_value	
TideWAL	4	3.21	TxExecutor::read_internal	
TideWAL	5	3.03	TxExecutor::mainte	
YCSB-C is read-only; WAL bytes and fdatsync counts are zero for all systems.				[12]

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